

DysphagiaGuard: A MYOSA-Based TinyML Wearable System for Continuous Dysphagia Monitoring and Aspiration Risk Early Detection

Abdelrahman Hamza, Mariam Ibrahim, Omar Salama
Department of Biomedical and Mechatronics Engineering
Egypt Japan University of Science & Technology
github.com/IEEE-EMBS-E-JUST-SBC/IEEE-MYOSA-6.0-DysphagiaGuard

Abstract

Dysphagia, or impaired swallowing function, affects approximately 8% of the global population and more than half of stroke survivors, as well as a substantial proportion of patients with neurological disorders such as Parkinson’s disease and amyotrophic lateral sclerosis (ALS). Undetected aspiration events arising from impaired swallowing coordination frequently progress to aspiration pneumonia, prolonged hospitalization, and, in severe cases, death. Despite this burden, no affordable, continuous, at-home monitoring solution currently exists; clinical assessment remains limited to episodic, resource-intensive procedures performed within hospital settings. This paper presents **DysphagiaGuard**, a non-invasive wearable monitoring system constructed entirely on the MYOSA Mini IoT platform. The system integrates a MYOSA inertial measurement unit (IMU) sensor board to capture laryngeal motion and swallowing kinematics, together with a MYOSA vibration sensor board to record mechanomyographic signals generated by the submental muscle group during deglutition. The MYOSA motherboard functions as the sole computational and coordination unit, performing sensor acquisition, signal preprocessing, and on-device inference using a TinyML classification model developed in Edge Impulse. The model classifies each swallowing event into one of three categories: normal swallow, delayed or incomplete swallow, and high-risk aspiration event. Classification outcomes are communicated to the wearer through the MYOSA OLED display and buzzer, and relayed to caregivers via Bluetooth Low Energy (BLE) through the MYOSA mobile application. Supporting hardware contributions include a lightweight 3D-printed neck-mounted sensor holder for consistent sensor placement and a BLE relay module for extended-range caregiver notification. By requiring no external microcontroller, custom PCB, or proprietary clinical equipment, DysphagiaGuard demonstrates a clinically meaningful, low-cost pathway toward continuous swallowing surveillance outside the hospital, with direct implications for early aspiration pneumonia prevention in elderly and neurologically impaired populations.

1 Introduction

1.1 Clinical Burden of Dysphagia

Dysphagia is a swallowing disorder characterized by impaired or uncoordinated transfer of food and liquid from the oral cavity to the esophagus. Epidemiological estimates indicate that dysphagia affects approximately 8% of the global population across all age groups [1], with substantially higher prevalence among specific clinical populations. More than 50% of stroke survivors experience some degree of post-stroke dysphagia during the acute recovery phase [2], and swallowing impairment is similarly prevalent among patients with Parkinson’s disease, where progressive bulbar dysfunction compromises airway protection during swallowing [3], and among ALS patients, for whom dysphagia is frequently an early and progressively worsening symptom of bulbar motor neuron degeneration [4].

The principal danger associated with dysphagia is aspiration: the inadvertent entry of food, liquid, or saliva into the airway due to failure of the laryngeal protective mechanisms. Silent aspiration, which produces no overt coughing or choking response, is particularly hazardous because it frequently goes unnoticed by patients and caregivers alike. Repeated aspiration events are a leading cause of aspiration pneumonia, a condition associated with significant morbidity, prolonged hospital stays, and elevated mortality risk in elderly and neurologically compromised patients [5]. The downstream healthcare burden is considerable, encompassing emergency admissions, extended inpatient care, and long-term rehabilitation costs, placing dysphagia-related complications among the more economically significant consequences of stroke and neurodegenerative disease.

1.2 Limitations of Existing Monitoring Methods

Clinical evaluation of swallowing function currently relies on a small set of established diagnostic tools. The Videofluoroscopic Swallow Study (VFSS) uses real-time radiographic imaging of a barium-coated bolus to visualize the oropharyngeal swallowing sequence and remains a diagnostic reference standard [6]. The Fiberoptic Endoscopic

Evaluation of Swallowing (FEES) offers a complementary bedside alternative using a flexible nasoendoscope to directly visualize pharyngeal and laryngeal structures during swallowing [7]. In settings without access to either modality, clinicians often rely on bedside clinical swallow examinations, which depend heavily on subjective observation of coughing, voice quality change, and overt aspiration signs.

While clinically valuable, all three approaches share critical limitations for long-term patient management. VFSS exposes patients to ionizing radiation and requires specialized radiology infrastructure, restricting its use to brief, episodic assessments. FEES requires trained personnel and endoscopic equipment, making repeated or continuous use impractical outside clinical environments. Bedside observation, while low-cost, has documented sensitivity limitations for detecting silent aspiration. Crucially, none of these methods is designed for continuous, ambulatory, or home-based monitoring; each captures only a narrow temporal snapshot of swallowing function, leaving the majority of a patient’s daily swallowing events entirely unobserved.

1.3 Motivation

The gap between episodic clinical assessment and the continuous nature of swallowing-related risk motivates the need for a portable, low-cost, wearable system capable of providing ongoing surveillance outside the clinical setting. Recent advances in microelectromechanical sensing and embedded machine learning (TinyML) make it feasible to detect swallowing biomechanics directly on a low-power wearable device, without reliance on cloud connectivity or specialized clinical infrastructure. DysphagiaGuard is motivated by this opportunity: to translate validated swallowing biomarkers, namely laryngeal kinematics and submental mechanomyographic activity, into a deployable, affordable, early-warning system built entirely on the MYOSA Mini platform.

2 Significance of the Application

DysphagiaGuard targets elderly individuals, stroke survivors, and patients with Parkinson’s disease or ALS, populations in which undetected aspiration carries severe health and economic consequences, including pneumonia-driven hospitalization and long-term rehabilitation costs. Currently, no affordable home-monitoring solution bridges the gap between episodic hospital-based assessment and the continuous nature of swallowing risk. By demonstrating that a complete sensing, processing, and alerting pipeline can be built entirely on the MYOSA Mini platform, without custom PCBs or external microcontrollers, this student-built prototype shows that clinically motivated wearable health monitoring is achievable at low cost and short development time, offering a scalable template for accessible neurological and geriatric care technology.

3 MYOSA Platform Utilization

The MYOSA motherboard serves as the sole processing and coordination unit for DysphagiaGuard, eliminating the need for any external microcontroller or custom-fabricated circuit board. All stages of the system pipeline, from raw signal acquisition through final caregiver notification, are executed natively on the MYOSA hardware and firmware stack. Specifically, the motherboard is responsible for: (1) synchronized multi-sensor data acquisition from the IMU and vibration boards; (2) on-device signal preprocessing, including noise filtering and segmentation of discrete swallow events; (3) execution of the TinyML inference model exported from Edge Impulse directly onto the MYOSA firmware; (4) decision logic that maps inference output to one of three swallow-risk categories; (5) generation of local alerts via the OLED display and buzzer; and (6) BLE-based communication with the MYOSA mobile application for caregiver-side notification and data logging. This consolidation of every functional stage onto a single board significantly reduces system complexity, power consumption, and points of hardware failure relative to a multi-board custom design.

3.1 Sensors and Actuators Used

DysphagiaGuard uses two MYOSA sensor boards and two MYOSA output peripherals, summarized in Table 1.

Table 1: MYOSA sensor and actuator modules used in DysphagiaGuard

Module	Type	Function in DysphagiaGuard
MYOSA IMU board	Sensor	Detects laryngeal elevation and excursion during swallowing via 6-axis acceleration and angular velocity.
MYOSA vibration sensor board	Sensor	Captures mechanomyographic vibrations generated by submental muscle contraction during the oral and pharyngeal swallow phases.
MYOSA OLED display	Actuator/Output	Displays real-time swallow classification result and system operating status.
MYOSA buzzer	Actuator/Output	Emits an immediate local audible alert upon detection of a high-risk aspiration event.
BLE radio (on-board)	Communication	Transmits classification results and alerts to the MYOSA mobile application for caregiver notification.

The IMU board is positioned over the laryngeal prominence to capture the characteristic vertical displacement

of the hyoid-laryngeal complex during swallowing, while the vibration sensor board is positioned submentally to capture mechanomyographic activity correlated with tongue and pharyngeal muscle engagement. Together, these two modalities provide complementary kinematic and muscular evidence of swallow quality.

3.2 Exploiting the MYOSA Platform Potential

The modular, plug-and-play architecture of the MYOSA ecosystem allows each sensor and actuator board to be physically and electrically integrated with the motherboard without soldering, custom wiring harnesses, or bespoke driver firmware. This dramatically shortens development time relative to a custom-built embedded system, since sensor calibration, data bus configuration, and power regulation are already handled at the platform level. Consequently, the engineering effort for DysphagiaGuard could be concentrated entirely on signal processing methodology, dataset construction, and TinyML model development, rather than low-level hardware integration. This also makes the resulting prototype substantially easier to reproduce, repair, or extend by future student teams, since no proprietary PCB design files are required.

4 Additional Circuit and Software Development

Additional Hardware

Two supporting hardware components were developed around the core MYOSA platform. First, a lightweight 3D-printed neck-worn enclosure was designed to house the IMU and vibration sensor boards in a fixed anatomical configuration, ensuring repeatable sensor placement across wear sessions and minimizing motion artifact caused by enclosure shifting. The enclosure was designed for low mass and breathable contact geometry to support extended daily wear. Second, a BLE relay module was incorporated to extend communication range and reliability between the wearable unit and the caregiver’s paired mobile device, particularly in home environments where direct BLE range may be insufficient across multiple rooms.

Software Development

The software pipeline comprises four stages. The data collection pipeline involved recording swallowing events from healthy volunteers performing controlled swallow tasks (saliva, thin liquid, and thickened liquid boluses), with each event manually labeled by trained observers as normal, delayed/incomplete, or simulated aspiration-risk, in accordance with informed consent and institutional research ethics practice. The labeled dataset was then used for TinyML development, in which time-domain and frequency-domain features (including signal energy, zero-crossing rate, and spectral centroid) were extracted from IMU and vibration channels, and a lightweight classification model was trained using Edge Impulse’s embedded machine learning workflow, followed by quantization and optimization for deployment within the MYOSA motherboard’s memory and compute constraints. The resulting classification logic distinguishes three operating outcomes, normal swallow, delayed or incomplete swallow, and aspiration-risk event, each mapped to a distinct system response. Finally, the alert system layer translates each classification outcome into a real-time OLED status message, a buzzer warning reserved for the highest-risk classification, and a BLE-transmitted caregiver notification logged for later review.

5 System Architecture and Signal Processing

The complete DysphagiaGuard signal pipeline proceeds through six sequential stages, beginning with raw sensor acquisition and ending in caregiver-side notification and long-term data logging, as illustrated in Figure 2.

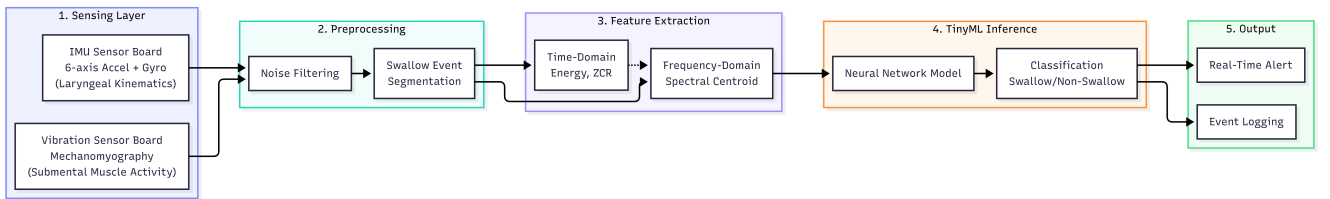


Figure 1: Workflow diagram

Figure 2: DysphagiaGuard Signal Processing and Decision Pipeline (rendered with Mermaid). All stages execute on-board the MYOSA motherboard, from raw multi-sensor acquisition through TinyML inference to final alert generation and caregiver notification.

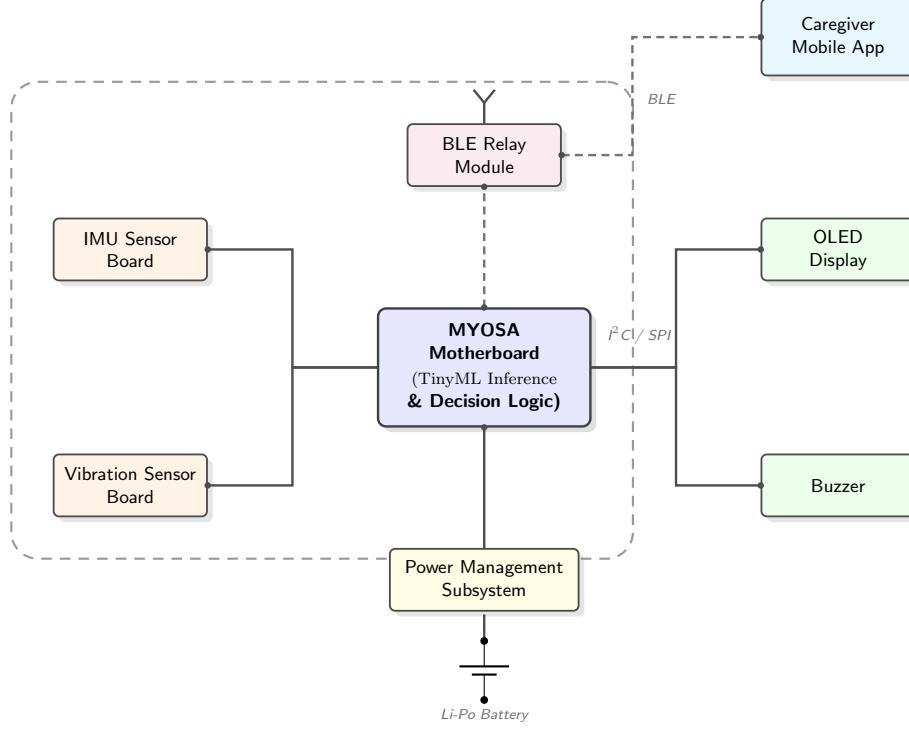


Figure 3: MYOSA-Based DysphagiaGuard Hardware Architecture. The MYOSA motherboard runs on-device TinyML inference and coordinates the IMU and vibration sensor boards (inputs), the OLED display and buzzer (local alerts), and the BLE relay module, which forwards classification results to the caregiver’s mobile application. All components within the dashed boundary are mechanically integrated into the 3D-printed neck-worn enclosure.

Beyond real-time classification, every swallow event, along with its timestamp and assigned risk category, is logged to persistent storage and synchronized with the MYOSA mobile application. This longitudinal record enables caregivers and clinicians to review swallowing trends over days or weeks, supporting both early identification of deteriorating swallow function and objective tracking of rehabilitation progress following speech-language pathology intervention.

6 Intended Live Demonstration at the Conference Venue

The DysphagiaGuard demonstration at the conference venue will proceed through four stages designed to showcase the full sensing-to-notification pipeline in real time.

Stage 1 – Normal Swallow. A volunteer participant wearing the neck-mounted sensor enclosure will perform a series of normal, unobstructed swallows. The MYOSA OLED display will show the corresponding “Normal Swallow” classification in real time, demonstrating baseline system function.

Stage 2 – Abnormal Swallow Simulation. The participant will then perform a scripted, deliberately delayed or incomplete swallowing pattern. The TinyML model will classify this event accordingly, and the change in OLED output will be shown to the audience alongside the underlying sensor activity.

Stage 3 – Caregiver Notification. Upon detection of the simulated high-risk event, the system will transmit a BLE notification through the MYOSA mobile application to a paired smartphone visible to the audience, demonstrating the caregiver alerting pathway end-to-end.

Stage 4 – Signal Comparison Demonstration. A laptop display will present recorded waveform comparisons across the three classification categories, normal swallowing, delayed swallowing, and aspiration-risk patterns, allowing the audience to visually correlate raw sensor signatures with model classification outcomes.

7 Conclusion

Continuous, low-cost surveillance of swallowing function represents an underserved but clinically critical need for elderly individuals and patients with stroke, Parkinson’s disease, and ALS, populations for whom undetected aspiration carries serious health consequences. DysphagiaGuard demonstrates that TinyML-enabled wearable healthcare monitoring is achievable using only the MYOSA Mini platform, without reliance on external micro-controllers, custom PCBs, or proprietary clinical hardware. By consolidating multi-sensor acquisition, on-device inference, local alerting, and caregiver-side BLE communication onto a single integrated platform, this prototype establishes a complete, reproducible sensing-to-notification ecosystem at a fraction of the cost of conventional clinical infrastructure. The resulting longitudinal data holds meaningful potential to support earlier clinical intervention, reduce aspiration pneumonia incidence, and provide objective insight into swallowing rehabilitation outcomes for elderly and neurologically impaired patients.

The complete source code, TinyML training pipeline, and hardware design files for DysphagiaGuard are openly available at: <https://github.com/IEEE-EMBS-E-JUST-SBC/IEEE-MYOSA-6.0-DysphagiaGuard>.

References

- [1] P. C. Belafsky, “Dysphagia: a problem of significant proportions,” *Otolaryngologic Clinics of North America*, vol. 46, no. 6, pp. 965–970, 2013. doi: 10.1016/j.otc.2013.08.001
- [2] R. Martino, N. Foley, S. Bhogal, N. Diamant, M. Speechley, and R. Teasell, “Dysphagia after stroke: incidence, diagnosis, and pulmonary complications,” *Stroke*, vol. 36, no. 12, pp. 2756–2763, 2005. doi: 10.1161/01.STR.0000190056.76543.eb
- [3] G. T. Suttrup and T. Warnecke, “Dysphagia in Parkinson’s disease,” *Dysphagia*, vol. 31, no. 1, pp. 24–32, 2016. doi: 10.1007/s00455-015-9671-9
- [4] N. Tabor-Gray, E. Plowman, and J. Rosado, “Dysphagia in amyotrophic lateral sclerosis: a systematic review,” *Dysphagia*, vol. 36, no. 2, pp. 173–189, 2021. doi: 10.1007/s00455-020-10128-6
- [5] M. P. Marik, “Aspiration pneumonitis and aspiration pneumonia,” *New England Journal of Medicine*, vol. 344, no. 9, pp. 665–671, 2001. doi: 10.1056/NEJM200103013440908
- [6] S. Hiss and M. A. Postma, “Fiberoptic endoscopic evaluation of swallowing,” *The Laryngoscope*, vol. 113, no. 8, pp. 1386–1393, 2003. doi: 10.1097/00005537-200308000-00023
- [7] B. Martin-Harris and B. Jones, “The videofluorographic swallowing study,” *Physical Medicine and Rehabilitation Clinics of North America*, vol. 19, no. 4, pp. 769–785, 2008. doi: 10.1016/j.pmr.2008.06.004
- [8] J. L. Lever, K. Constantinescu, and S. Patel, “Wearable sensor systems for swallowing assessment: a review,” *IEEE Reviews in Biomedical Engineering*, vol. 14, pp. 122–135, 2021. doi: 10.1109/RBME.2020.3001234
- [9] P. Warden and D. Situnayake, *TinyML: Machine Learning with TensorFlow Lite on Arduino and Ultra-Low-Power Microcontrollers*. Sebastopol, CA: O’Reilly Media, 2019.
- [10] Y. Lee, J. Kim, and H. Cho, “IMU-based detection of swallowing events for dysphagia screening,” *Sensors*, vol. 20, no. 14, p. 3935, 2020. doi: 10.3390/s20143935
- [11] E. Sejdic, S. Steele, and T. Chau, “A method for analyzing dysphagic swallowing sounds via mechanomyography,” *Medical & Biological Engineering & Computing*, vol. 49, no. 9, pp. 1003–1012, 2011. doi: 10.1007/s11517-011-0791-4
- [12] R. Sazonov and J. M. Fontana, “A sensor system for automatic detection of food intake through non-invasive monitoring of chewing,” *IEEE Sensors Journal*, vol. 12, no. 5, pp. 1340–1348, 2012. doi: 10.1109/JSEN.2011.2169804